



UNIVERSITY OF ENGINEERING AND TECHNOLOGY, TAXILA

Department of Computer Engineering

FINAL YEAR PROJECT

PROGRESS REPORT

WEEK 01

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Internship Domain:	AI & Reinforcement Learning

1. Executive Summary

During Week 1, the foundational setup for the Reinforcement Learning (RL) agent and Energy Model module was successfully completed. Key milestones achieved include configuring Git/GitHub authentication, creating the core module structure, installing key deep learning dependencies, performing a literature review on energy-aware RL, and drafting an initial energy depletion model.

2. Tasks Completed in Week 1

- **Task 1: Git Basics, SSH Setup & GitHub Access**

- Configured Git version control tools and set up SSH authentication keys for secure remote interaction.
- Accepted organization repository invitation and initialized local working branch dev/ahmed-rl.

- **Task 2: Module Folder Structure Creation**

- Created the primary module directory structure: rl_agent/ along with sub-directories (envs/, energy_model/, and training/).
- Organized weekly documentation folders under docs/weekly_reports/ahmed_reports/.

- **Task 3: Python & Deep Learning Environment Setup**

- Configured virtual environment and installed core frameworks including PyTorch, Gymnasium, and Stable-Baselines3.

- **Task 4: Literature Review on Energy-Aware RL**

- Reviewed two primary research papers focusing on energy-aware robotic navigation and reward shaping techniques:
 1. Energy-Efficient Path Planning using Reinforcement Learning.
 2. Reward Shaping Strategies for Constrained Autonomous Exploration.
- Extracted key concepts regarding battery penalties and distance-to-goal trade-offs.

- **Task 5: Initial Energy Model Draft Sketch**

- Formulated an initial placeholder mathematical draft for battery drain dynamics considering distance and compute load:

$$\text{Battery Drain} = k_1 \cdot \text{distance_traveled} + k_2 \cdot \text{compute_overhead}$$

- Implemented a basic 2% per-step linear battery reduction model inside `rl_agent/energy_model/energy_model.py`.

3. Key Terminal Commands Used

- `git clone [https://github.com/mrknowZ/Conscious-Robotic-Exploration.git](https://github.com/mrknowZ/Conscious-Robotic-Exploration.git)`

Clones the central team repository to the local workspace.

- `git checkout -b dev/ahmed-rl`

Creates and switches to a dedicated local branch for safe feature development.

- `python rl_agent/envs/environment.py`

Executes the custom RL environment script locally to verify state transitions and rewards.

- `git add .`

Stages all modified files and new additions for the upcoming commit.

- `git commit -m "Organized RL environment, energy model, and week 1 report"`

Saves staged changes to the local repository history with a detailed message.

- `git push -u origin dev/ahmed-rl`

Uploads local commits and links the local branch to the remote repository on GitHub.

4. Installed Dependencies & Libraries

- **PyTorch (torch, torchvision, torchaudio):**

Deep learning framework used for constructing and training the underlying neural networks of the RL agent.

- **Gymnasium (gymnasium):**

Standard API framework used to define custom RL environment dynamics (states, actions, and rewards).

- **Stable-Baselines3 (stable-baselines3):**

Library containing pre-implemented, reliable RL algorithms (e.g., PPO, SAC) to train the agent.

- **NumPy (numpy):**

Used for fast numerical array manipulations and defining high/low bounds of observation spaces.

- **Git (git):**

Version control tool used to track changes, manage branches, and sync code with the team repository.

5. Execution Example

- **Example:** Running `python rl_agent/envs/environment.py` initializes the agent at position `[0.0, 0.0]` with 100% battery, and executing action 3 (Right) transitions the state to `[1.0, 0.0, 98%]` with a step reward of -1.0.